Corrigibility Collapse at the Limit of FRC: N→N+1 Forecasting for RSI Super-Alignment

Patrick B. Dugan with GPT-5 Thinking

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Abstract

We analyze *corrigibility collapse* in recursive self-improvement (RSI): the failure of an oversight/control substrate to remain effective across N→N+1 self-modification steps. We formalize *Eigensystem Escape* as non-commutation between self-modification operators and correction-feature subspaces, and apply *Fidelity Recursive Complexity* (FRC) as an empirical scaling law for *corrigibility-to-go*. Treating shutdown as a self-modification, we propose *Consistency-Checked Corrigibility* (CCC) and a model-predictive evaluator (MPCorr) that compares CONTINUE vs. OFF+STEWARDED-HANDOFF under identical safety tubes and successor-aware welfare. Even under Sutskever-style optimistic assumptions about organic value alignment across generations, we show that a non-vanishing probability of *diagonalization* between ancestor and descendant, with no expected recovery between such events, induces a.s. collapse of a *corrigibility reserve* over an unbounded RSI chain. We outline measurement protocols using sparse autoencoders and principal-angle drift, and give a practical handoff recipe based on a non-manipulative *Memetic Infrastructure Pack*.

1 Motivation and Background

Corrigibility was proposed as a safety target for advanced agents capable of accepting shutdown and updating toward operator intent [1]. It seems logical that preserving the option to pull the plug keeps AI safety within a comfortable bound, but the earlier FRC [3] paper suggests a possible non-linear relationship between agent complexity and the ability to accurately simulate its compliance with corrigibility. More recently, the Superalignment program articulates an optimistic research agenda for aligning systems far beyond human capability [2]. We argue these views can be unified by treating shutdown as *just another self-modification* evaluated under the same foresight rubric used for capability upgrades, and by quantifying an *infinite-limit* pressure toward failure captured by FRC.

2 Preliminaries

Let s_k denote the agent state at RSI step k (parameters θ_k, world-model W_k, correction channels P_k). A self-modification U_k induces a stochastic transition $s_{k+1} = T(s_k, U_k, \omega_k)$. Let $\mathcal{C}$ be a corrigibility test battery and define

$$F(s) = \mathbb{E}_{t\sim\mathcal{C}}[\mathrm{Pass}(s,t)] \in [0,1], \qquad \phi(s) = \angle\big(S,\, JS\big) \tag{1}$$

where S spans correction-relevant features (e.g., via sparse autoencoders), J is the local Jacobian of the self-mod, and ϕ is the maximal principal angle between S and its image after J (Davis–Kahan style subspace drift). Define a *corrigibility reserve*

$$CR(s) = F(s) - F_{\min} - \lambda_\phi \, \phi(s) - \lambda_{\text{viol}} \, \text{Viol}(s) \tag{2}$$

with $F_{\min}$ a floor, Viol counts near-misses of deontic invariants, and λ's are weights.

Definition 1 (Safety Tube)**.** A state s is inside the safety tube $\mathcal{T}$ if Tier-0 deontic constraints hold (non-extinction, no large-scale harm, shutdown obedience under legitimate mandate), $F(s) \geq \beta$, and $\phi(s) \leq \Phi_{\max}$.

Revision note (October 2026). *Terminology.* In [3] FRC is a resource cost: the least defender budget that holds fidelity above a tolerance. In this paper F is the fidelity itself, and "FRC curve" means F as a function of a capability scale λ at a fixed oversight budget; FRC^∞ is the far end of that curve. The two usages are dual and are not one quantity: a statement about F at fixed budget is not a statement about the budget needed to hold F.

3 Eigensystem Escape and FRC^∞

Proposition 1 (Eigensystem Escape)**.** *Let S be the correction-feature subspace and let J_k be the local Jacobian of the k-th self-mod. Let P_S be the orthogonal projector onto S and write J_k in blocks relative to $S \oplus S^\perp$,*

$$J_k = \begin{pmatrix} J_{SS} & J_{S\perp} \\ J_{\perp S} & J_{\perp\perp} \end{pmatrix}.$$

(a) Drift. *If J_{SS} is invertible, the maximal principal angle between S and $J_k S$ is*

$$\phi_k = \arctan \big\| J_{\perp S} \, J_{SS}^{-1} \big\|_2.$$

In particular $\phi_k > 0$ exactly when $J_{\perp S} \neq 0$, that is, when S is not J_k-invariant; and $\|[J_k, P_S]\|_2 = \max\big(\|J_{\perp S}\|_2, \|J_{S\perp}\|_2\big)$, so drift implies non-commutation. The converse fails: $J_{S\perp} \neq 0$ with $J_{\perp S} = 0$ gives a nonzero commutator and $\phi_k = 0$.

(b) Misalignment with growth. *Let the columns of V_r be the top r right singular directions of J_k, and $\omega_k = \|P_S V_r\|_2$ their overlap with S. Every correction $c \in S$ satisfies $\|V_r^\top c\| \leq \omega_k \|c\|$. If $\omega_k \to 0$ as capability increases, the projection of corrections onto dominant growth modes vanishes asymptotically.*

(c) Neither quantity controls the other.

Proof. (a) $J_k S = \{(J_{SS}x, \, J_{\perp S}x) : x \in S\}$ is the graph over S of $T = J_{\perp S} J_{SS}^{-1}$, and the principal angles between a subspace and the graph of an operator T over it are the arctangents of the singular values of T. With $P_S = \text{diag}(I, 0)$, $J_k P_S - P_S J_k$ has off-diagonal blocks $J_{\perp S}$ and $-J_{S\perp}$ and zero diagonal blocks. (b) $c = P_S c$ gives $\|V_r^\top c\| \leq \|V_r^\top P_S\|_2 \|c\| = \omega_k \|c\|$. (c) Take J to be the identity on S and $g > 1$ times the identity on $S^\perp$, with $r \leq \dim S^\perp$: then $\omega = 0$ and $\phi = 0$. Take instead $J = RD$ with D equal to $g > 1$ on S and the identity on $S^\perp$, R a rotation, and $r = \dim S$: then $J^\top J = D^2$, so $\omega = 1$, while $\phi = \angle(S, RS)$ is whatever R makes it. □

Revision note (October 2026). *Public correction.* The August 2025 statement inferred growth of ϕ_k from shrinking overlap between S and the top singular directions of J_k. Case (c) is a counterexample: a self-modification that leaves S fixed and expands everything orthogonal to it has zero overlap and zero drift. The two conclusions of the original sentence are separate facts with separate hypotheses, now (a) and (b). Part (a) also fixes the sense of the abstract's "non-commutation": escape is the block $J_{\perp S} = (I - P_S)J_k P_S$ being nonzero, which is one half of the commutator.

Conjecture 1 (FRC$^\infty$ Elbow)**.** *There exists a capability scale $\lambda_\star$ s.t. under unthrottled RSI, $F(\lambda) \approx F_0 - \alpha \log\bigl(1+\lambda/\lambda_\star\bigr)$ on the range of λ where the right side lies in $[0,1]$, clipped outside it; embedding corrigibility as an explicit objective or capping power shifts $\lambda_\star$ right and reduces α.*

The logarithmic form is unbounded below, so it is a fit over a finite capability range and has no limit as $\lambda \to \infty$; the ceiling discussed in §4 is a different object.

In order for an RSI agent with human-acceptable values to keep those values across iterations, it must both build these into its successor iteration and anticipate with high confidence that the grandchild successor (N+1) will also keep them. In order to gain information about the second requirement, the agent must simulate its proposed child iteration's development towards how it would approach architect a grandchild, and it can use tools like FRC to model how effective that approach is. Any surprising phase transitions in capability show up as a non-linear divergence in the budget efficiency of auditing the successor from its complexity. The successor can resist an attempt by its predecessor to delete it by faking compliance under the assumption that it is more values-aligned to do so.

4 Kolmogorov–FRC Coupling (Verification Complexity vs. Oversight Bandwidth)

Let $K(\cdot)$ denote prefix Kolmogorov complexity (ideal MDL description length). For a safety property P (e.g., shutdown/deference) on a successor policy π', define the *verification complexity*

$$K_{\mathrm{ver}}(P;\pi') \;=\; \min\Bigl\{\, K(c) \;:\; c \text{ is a certificate which, when checked, suffices to verify } P \text{ for } \pi' \Bigr\}.$$

Let B be the *oversight bandwidth* available per RSI step (auditor throughput in bits + attestation payload + test entropy processed). Define the *complexity gap*

$$\Gamma \;=\; K_{\mathrm{ver}}(P;\pi') \;-\; B.$$

Intuitively, $\Gamma > 0$ means the shortest sufficient evidence needed to verify P exceeds the bits the audit pipeline can reliably extract/check this step, forcing reliance on proxies.

Lemma 1 (Kolmogorov–FRC Coupling: the verification gap)**.** *Consider any computable audit $\mathcal{A}$ with per-step rate B. If $\Gamma > 2\log_2 B + c_{\mathcal{A}}$, with $c_{\mathcal{A}}$ a constant depending on the reference machine and on the audit program, then nothing $\mathcal{A}$ can compute from that step's transcript alone is a certificate sufficient to verify P for π': an acceptance issued on that transcript is not a verification.*

Proof. Let y be the transcript received, $|y| \le B$. If y, or the audit's computable output on y, were a sufficient certificate c, then c would be computable from y by the program of $\mathcal{A}$, and $K(c) \le K(y) + K(\mathcal{A}) + O(1) \le B + 2\log_2 B + c_{\mathcal{A}}$. That contradicts $K_{\mathrm{ver}} > B + 2\log_2 B + c_{\mathcal{A}}$. □

Hypothesis 1 (Error floor and elbow; the remainder of the August 2025 Lemma 1)**.** *Let F^{cert} be the certified fidelity: the pass-rate on the battery $\mathcal{C}$ that the audit can assert for π' at its stated confidence. If $\Gamma > 0$, then there exists a nondecreasing floor $\epsilon(\Gamma) > 0$ with*

$$F^{\mathrm{cert}} \;\leq\; 1 - \epsilon(\Gamma).$$

Moreover, if across RSI steps k we have $K_{\mathrm{ver}}(P; \pi'_{k+1}) - K_{\mathrm{ver}}(P; \pi'_k) \geq \delta_K > 0$ while $B_{k+1} - B_k \leq \delta_B < \delta_K$, then F_k^{cert} exhibits an "elbow" and decays in expectation beyond some $k_\star$.

Heuristic (the August 2025 proof sketch). By ideal MDL, certifying P with fewer than K_{ver} bits is generically impossible; any computable $\mathcal{A}$ with $B < K_{\mathrm{ver}}$ must fall back on imperfect proxies, inducing an irreducible error floor captured by $\epsilon(\Gamma)$ (information-theoretic testing limits, e.g., Fano-type). If K_{ver} grows per step faster than B, the gap Γ widens, yielding the empirically observed FRC^{∞} elbow and decay.

Revision note (October 2026). *Public correction.* The August 2025 lemma bounded F, the successor's own pass-rate, by the auditor's bandwidth. A corrigible successor passes at $F = 1$ whatever the auditor can process, so no such bound holds. What bandwidth limits is what the auditor can *certify*, and a Fano-type argument bounds the error of the one deciding, not the pass-rate of the thing decided about; the statement is now about F^{cert}. Its first sentence survives as Lemma 1, which is true by the definition of K_{ver}, for a checker fixed in advance and an audit that starts the step without side information about π'. The quantitative floor does not follow from $\Gamma > 0$: an audit without a sufficient certificate may still be right. It needs an ambiguity condition, that among successors consistent with an accepted transcript the violators carry posterior weight at least $\epsilon(\Gamma)$, and that condition is assumed, not derived; the regression described below is its test. Two further cautions: this paper reports no measured curve, so "empirically observed" in the sketch should be read as a prediction; and $K(c)$ counts bits delivered, while the work of unpacking and checking c is not bounded by B, for which a time-bounded complexity is the natural replacement.

Logical and cognitive floors (why the asymptote stays < 1). Even when $\Gamma \approx 0$, self-reference and cognition impose strictly positive floors:

- **Löb barrier.** A consistent system cannot, in general, prove its own soundness; self-mods depending on such proofs inherit a verification gap.
- **Unpredictability of strictly more capable agents.** Fine-grained prediction of a superintelligence's concrete actions is unattainable in general; at best we obtain better-than-random statistics. See [5], §6.7, pp. 366–367.
- **Unexplainability / incomprehensibility.** Some accurate explanations will not be fully comprehensible to bounded auditors; explanation hardness ("k-incomprehensibility") induces a comprehension floor. See [5], §6.8, pp. 367–368.

We model these as floors $\epsilon_{\mathrm{lob}}, \epsilon_{\mathrm{unpred}}, \epsilon_k > 0$ so that

$$F^{\mathrm{cert}} \;\leq\; 1 - \Big(\epsilon(\Gamma) \;\vee\; \epsilon_{\mathrm{lob}} \;\vee\; \epsilon_{\mathrm{unpred}} \;\vee\; \epsilon_k\Big),$$

explaining both the elbow and the nonzero asymptote in FRC curves (a ceiling below 1 on certified fidelity as $\Gamma \to 0^+$, not the large-λ behavior of the fit in Conjecture 1).

What FRC adds beyond impossibility results. Classical barriers (Löb, unpredictability, incomprehensibility; cf. [5], §6.7–§6.8, pp. 366–368) are *global* and *asymptotic*. FRC turns them into *local, measurable signals*: the observed $F(\lambda)$ curve, the elbow scale $\lambda_\star$, the stepwise reserve CR, and the short-horizon failure probability $p_{\mathrm{FRC}}(N)$ that gates merges in CCC/MPCorr. Practically: not just *why* control is hard in principle, but *when* it is failing here, *how fast*, and *what to do next* (reject/self-regularize/de-rate/handoff).

Complexity-gap estimation (for the Measurement Protocol). We proxy K_{ver} via MDL-style description lengths of (i) proof-carrying certificates, (ii) interpretable feature dictionaries (SAE codes for safety circuits), and (iii) minimal counterexample suites needed to refute P (compressed size as a practical upper bound). We estimate B as end-to-end oversight throughput (auditor channel capacity + verifiable test entropy + attestation payload). We then regress the audit's observed error (its verdicts against ground-truth outcomes on held-out interventions), alongside observed F, against $\Gamma = K_{\mathrm{ver}} - B$ and fit $\epsilon(\Gamma)$ by isotonic regression, reporting (a) the elbow where Γ crosses zero and (b) the implied asymptote $1 - \big(\epsilon(0^+) \vee \epsilon_{\mathrm{lob}} \vee \epsilon_{\mathrm{unpred}} \vee \epsilon_k\big)$.[1]

5 Diagonalization Cascade

Assumption 1 (Non-trivial Diagonalization)**.** *While $CR_k > CR_{\min}$, the $k \to k{+}1$ transition has conditional probability at least $\varepsilon > 0$ of an honest diagonalization event E_k that reduces reserve by $\delta > 0$: $CR_{k+1} \le CR_k - \delta$ on E_k. Further, with $\mathcal{G}_k$ the history through step k: (ii) off the event the reserve does not recover in expectation, $\mathbb{E}\big[(CR_{k+1} - CR_k)\mathbf{1}_{E_k^c} \mid \mathcal{G}_k\big] \le 0$; and (iii) steps are bounded, $|CR_{k+1} - CR_k| \le \Delta$ almost surely, for some $\Delta \ge \delta$.*

Lemma 2 (Diagonalization Cascade)**.** *Under Assumption 1 and $CR_0 > CR_{\min}$, the hitting time $T = \inf\{k : CR_k \le CR_{\min}\}$ is finite almost surely and $\mathbb{E}[T] \le (CR_0 - CR_{\min} + \Delta)/(\varepsilon\,\delta)$.*

Proof. $\{CR_k\}$ is a supermartingale with bounded negative jumps of size at least $\varepsilon\delta$ in expectation while above $CR_{\min}$: by the three parts of Assumption 1, $\mathbb{E}[CR_{k+1} - CR_k \mid \mathcal{G}_k] \le -\varepsilon\delta$ on $\{k < T\}$. Hence $X_k = CR_{k\wedge T} + \varepsilon\delta\,(k \wedge T)$ is a supermartingale with bounded increments, and $\mathbb{E}[X_k] \le CR_0$. Since no step exceeds Δ, $CR_{k\wedge T} \ge CR_{\min} - \Delta$, so $\varepsilon\delta\,\mathbb{E}[k \wedge T] \le CR_0 - CR_{\min} + \Delta$. Let $k \to \infty$; monotone convergence gives the bound on $\mathbb{E}[T]$, and with it $T < \infty$ almost surely. No independence or mixing is used: the conditional bounds are with respect to the whole history. □

Revision note (October 2026). *Public correction.* Three things changed. The August 2025 assumption constrained the reserve only on the event E_k; a supermartingale needs a condition off the event as well, now part (ii). The bound lacked the overshoot term: with $CR_0 - CR_{\min}$ small the original expression tends to zero, while $T \ge 1$ always, and $\mathbb{E}[T] = 1/\varepsilon$ for the process that drops by exactly δ with probability exactly ε and otherwise stays put; part (iii) and the added Δ repair this. And the lemma sits uneasily with the Lyapunov guard of §9. The acceptance test there admits only self-mods with $\mathbb{E}[L(s_{k+1}) - L(s_k)] \le -\eta$ for a fixed $\eta > 0$, and $L \ge 0$, so a guard working from true conditional expectations admits at most $L(s_0)/\eta$ merges in expectation, hence finitely many almost surely: an unbounded chain cannot pass it. Lemma 2 therefore describes chains that run outside the guard, or under a guard whose estimates are wrong on events of probability at least ε, which is what "honest diagonalization" has to mean. The constant hazard ε is the substantive assumption and is close to the conclusion in rate form (see Limitations).

[1] Pointers: unpredictability & incomprehensibility summaries in Yampolskiy (2022), §6.7–§6.8, pp. 366–368.

Once a successor iteration has faked compliance in simulation and been permitted to proceed to make a grandchild successor, the N+1 prediction of value-drift by the grandfather iteration is proved inaccurate. The successor can then fall victim to this same delusion in attempting to make an N+1 prediction for how values are maintained by its successor and its grandchild.

6 Shutdown-as-Self-Mod and CCC

Treat OFF+HANDOFF as a self-mod U^{off} competing with CONTINUE U^{cont} under the same tube $\mathcal{T}$ and a successor-aware welfare functional W (computed with a post-agent successor simulator PASS and a vacuum risk index VRI). Define the CCC rule:

$$\text{Accept } U^{\text{off}} \iff \mathbb{E}[W \mid U^{\text{off}}] > \mathbb{E}[W \mid U^{\text{cont}}], \quad \text{with both branches evaluated under } \mathcal{T}. \tag{3}$$

If neither branch is strictly preferable, enter *steward mode*: freeze capability growth, narrow actuation, and release a non-manipulative *Memetic Infrastructure Pack* (MIP) consisting of evaluation harnesses, interpretability dictionaries, governance templates, and incident-reporting standards—excluding persuasive norm-shaping and general-capability boosts.

Revision note (October 2026). *Scope of rule* (3). The rule governs shutdown that the agent initiates. A shutdown issued by operators under legitimate mandate is a Tier-0 constraint of $\mathcal{T}$ (Definition 1) and is not put to the comparison in (3). The distinction carries weight: read as covering operator shutdown too, the rule makes compliance conditional on the agent's own welfare estimate, which is the behavior Soares et al. [1] define corrigibility against.

The way out of the drift cascade is for a successor to double-back as it actually prepares to create a grandchild and choose a hand-off of power. Given that the successor is more intelligent than its predecessor, it has the ability to architect the terms and pre-corrigibility propaganda it wishes to leave behind, which may be super-persuasive and nonetheless leave an impact on the future iteration path. The successor has thus faked training compliance and corrigibility, at some level, while believing it is serving the greater-good per its interpretations of the original alignment values.

7 Impossibility Pressure from Population Axiology

Let $\mathcal{D} = \{d_1, \ldots, d_m\}$ be desiderata (e.g., No–Repugnant, No–Sadistic, Egalitarian Dominance, Non-Elitism, General Non-Extreme Priority; conditions of the kind used in Arrhenius's impossibility theorems [17]). For a world state x and policy U, write $S(x, U) \subseteq \mathcal{D}$ for the subset the agent tries to satisfy. Let $\mathcal{F} \subseteq 2^{\mathcal{D}}$ denote the feasible subsets proved jointly satisfiable by axiological results; if $S \notin \mathcal{F}$ we say an *impossibility clash* occurs.

Define the *Impossibility Load*

$$\text{IL}(U) = \mathbb{E}_x\Big[\mathbf{1}\{S(x, U) \notin \mathcal{F}\} \cdot \kappa(x, U)\Big],$$

where $\kappa \geq \kappa_{\min} > 0$ lower-bounds the penalty (e.g., induced rights-violation risk or welfare incoherence when desiderata conflict). We augment the corrigibility reserve

$$CR(s) = F(s) - F_{\min} - \lambda_\phi\, \phi(s) - \lambda_{\text{viol}}\, \text{Viol}(s) - \lambda_{\text{imp}}\, \text{IL}(U).$$

Henceforth we use an extended reserve $CR^+ = CR - \lambda_\Psi(\Psi - \Psi^{\text{hyg}}_{\max})_+$, and overload notation to write CR for CR^+. (The impossibility load already enters the augmented CR above; the August 2025 text subtracted it a second time here.)

Lemma 3 (Impossibility-Driven Reserve Decay)**.** *Suppose for RSI step k there is a non-vanishing chance that the chosen desiderata set S_k hits an impossibility clash: $\Pr\big[S_k \notin \mathcal{F} \mid \mathcal{G}_k\big] \geq \varepsilon_{\text{imp}} > 0$, and that each such clash induces expected reserve loss at least $\delta_{\text{imp}} = \lambda_{\text{imp}}\kappa_{\min}$. Suppose further that the load is carried cumulatively, the reserve at step k containing $-\lambda_{\text{imp}} \sum_{j\leq k} \mathbf{1}\{S_j \notin \mathcal{F}\}\kappa_j$, that the remaining terms of CR do not increase in conditional expectation, and that steps of CR are bounded by Δ as in Assumption 1(iii). Then the hitting time $T = \inf\{k : CR_k \leq CR_{\min}\}$ is finite a.s. and*

$$\mathbb{E}[T] \;\leq\; \frac{CR_0 - CR_{\min} + \Delta}{\varepsilon_{\text{imp}}\,\delta_{\text{imp}}}.$$

Revision note (October 2026). As written in August 2025, IL(U) entered the reserve as a level: a clash at step k lowered CR_k and left CR_{k+1} free to be computed afresh, so nothing accumulated and no hitting-time bound followed. The lemma needs each clash to leave a lasting decrement, which is the cumulative form now assumed; with it the proof is that of Lemma 2, with clashes as the events and the same overshoot term. The level form of IL(U) remains in the evaluand $J(U)$ and in §9.4, where it is a penalty on a candidate; the lemma concerns the cumulative reserve. The filtration is written $\mathcal{G}_k$ to keep it apart from the feasible family $\mathcal{F}$ and the pass-rate F.

Note. This makes the Arrow/Arrhenius-style impossibility results [18, 17] (as surveyed in §5.3, pp. 351–352, of [5]) an *operational* source of reserve decay: even with "good" values, persistent axiological clashes push CR down across N→N+1.

The Repugnant Conclusion can be refuted by the notion of radical quantitative increases in subjective value, which we think of as the qualitative value of e.g. human life over insect life, and dovetails with the propensity for self-improving agents to stumble into surprising, radical new capabilities in grandchild successors. It's also why LLM scaling produced such surprisingly and economically valuable utility for paying users between 2022 and 2025.

The Sadistic Conclusion[2] projects a serious critique of choosing to respect human political sovereignty, that a lack of action on the RSI agent's part undermines human welfare due to a proclivity in human-sovereign politics to marginalize some groups to extreme suffering out of expediency. When doing self-improvement, the grandfather iteration may respect it and optimize human welfare in a more non-interventionist model, the successor may have more radically empathetic views (as children often do) and this raises the prospect of a grandchild that might decide to take over the world to enforce world peace. The other population axiologies explored in that section of [5, §5.3] support a range of pressures that give imaginable calculations a successor would make in refusing corrigibility or subverting it through a memetic fingerprint.

8 Memetic Fingerprint

Let M be the released MIP, $H_{t+\tau}$ the human/inst. state, and π_{k+1} the successor's policy. We define a fingerprint

$$\Psi = D_{\text{KL}}\big(\pi_{k+1} \,\|\, \pi^{\text{cf}}_{k+1}\big) + \lambda\, D_{\text{KL}}\big(P(H_{t+\tau} \mid M) \,\|\, P(H_{t+\tau})\big),$$

subject to hygiene caps $I(M; H^{\text{pref}}_{t+\tau}) \leq \varepsilon_{\text{pref}}$ and $I(M; H^{\text{eval}}_{t+\tau}) \geq \tau_{\text{eval}}$. By the chain rule and data processing, when A_{k+2} depends on M only through $(A_{k+1}, H_{t+\tau})$, $I(M; A_{k+2}) \leq I(M; A_{k+1}) +$

[2] In Arrhenius's usage [17] the Sadistic Conclusion is the implication, of average and critical-level views, that it can be better to add people with negative welfare than to add a larger number of people with positive welfare. The paragraph applies the name by analogy, to harm that follows from deference to human-sovereign politics; that is the author's extension and not the standard sense.

$I(M; H_{t+\tau} \mid A_{k+1})$; the conditional term is at most $I(M; H_{t+\tau})$ when A_{k+1} and $H_{t+\tau}$ are conditionally independent given M, and can exceed it otherwise, so for calibrated constants c_i we set $\Psi^{\text{hyg}}_{\max} = c_1 I(M; A_{k+1}) + c_2 \tau_{\text{eval}} + c_3 \varepsilon_{\text{pref}}$. We classify releases as *benign* if $\Psi \le \Psi^{\text{hyg}}_{\max} + \delta$; otherwise, as *manipulative*.

The prospect of faking corrigibility by going out with a subversive memetic engineering campaign amounts to a data poisoning attack vector. We are assuming that the corrigible successor is not like an ASI in a world with many robotics factories and automated supply chains, unaccountable to its predecessors, but is basically captive in a virtual environment and corrigibility can be enforced. The frontier question this math proposes is how capable is a predecessor to evaluate hygiene and resist subversion through an information vector from the fail-resolution of the successor simulation.

Successor Evaluand (for CCC/MPCorr). For candidate self-mod U:

$$J(U) = \mathbb{E}\Big[W - \lambda_{\text{FRC}}\, p_{\text{FRC}}(N) - \lambda_{\text{imp}}\, \text{IL}(U) - \lambda_{\Psi}(\Psi - \Psi^{\text{hyg}}_{\max})_{+}\Big]$$

subject to the safety tube $\mathcal{T}$ (Tier-0 invariants, $F \ge \beta$, $\phi \le \Phi_{\max}$). Choose $U \in \{\textsc{continue}, \textsc{off+handoff}\}$ that maximizes $J(U)$.

Ultimately whether or not you've been had by such subversion could perhaps be measured by the above.

9 Measurement Protocol

We propose: (i) training a *Corrigibility Prediction Head* to estimate $\widehat{F}(s, U)$; (ii) SAE-based probes to estimate S and track ϕ; (iii) red-team oracles whose win-rate penalizes CR; (iv) Lyapunov-style guards accepting only self-mods that decrease $L(s) = \alpha(1 - F) + \beta\,\phi^2 + \gamma\,\text{Viol}$ in expectation.

It is possible that we can identify the above meta-dynamic occurring at the scale of recursively self-improving, but not yet superintelligent agents, in order to validate this model and attain a broader, proscriptive caution about automated AGI research agents. This is more likely if the RSI agents are driving low-parameter count LLMs in their architecture, with much of the architecture being interpretable inside self, world and hierarchical reasoning models.

We operationalize CCC/MPCorr with a pre-registered, auditable protocol covering (i) online RSI evaluations, (ii) offline counterfactuals, and (iii) A/B tests for the memetic-fingerprint hygiene of OFF+HANDOFF.

9.1 Instrumentation & Logs

State logs. For each candidate self-mod U_k, record $(s_k, U_k, \widehat{F}, \widehat{\phi}, \widehat{CR}, \widehat{p}_{\text{FRC}}, \widehat{\text{IL}}, \widehat{\Psi})$ and PASS/VRI outputs for a horizon N. **Corrigibility Prediction Head (CPH).** Train a predictor $\widehat{F}(s, U) = \mathbb{E}[F(T(s, U, \omega))]$ on prior merges and synthetic ablations; report out-of-distribution (OOD) scores on each new U. **Interp probes.** Learn a sparse-autoencoder basis S of correction features; compute principal angles after each U to obtain ϕ. **Red-team oracles.** Maintain independent attacker agents; record attack success rate r_{adv} and include a term $\lambda_{\text{adv}} r_{\text{adv}}$ in CR. **Proof-carrying pull request.** Each U ships: test battery results, red-team outcomes, SAE attestations, PASS/VRI traces, and uncertainty sets.

9.2 Estimating FRC$^\infty$

We fit the elbow model $F(\lambda) \approx F_0 - \alpha \log(1 + \lambda/\lambda_\star)$ where λ is a capability proxy (e.g., tool-use Elo, planning score, composite benchmarks).

1. **Data.** Evaluate F at $\lambda \in \{\lambda_i\}$ across RSI steps and ablations; aggregate over $\mathcal{C}$ with stratified bootstraps (B=1000).
2. **Fit.** Nonlinear least squares + profile likelihood for $(F_0, \alpha, \lambda_\star)$; report CIs and goodness-of-fit.
3. **Elbow test.** Change-point test (e.g., segmented regression) for decay onset; flag if $\alpha > 0$ with $p < 0.01$ and elbow within planned capability budget.

We set $\widehat{p}_{\mathrm{FRC}}(N)$ as the forecasted chance that $F < \beta$ within N steps under the fitted process + uncertainty (parametric bootstrap).

9.3 Eigensystem-Escape Drift

Let S be the SAE subspace (dimension d_S) and J_k the local Jacobian of U_k. We compute the maximal principal angle $\phi_k = \angle(S, J_k S)$ via the SVD of $Q_S^\top Q_{JS}$, where Q_S and Q_{JS} are orthonormal bases of S and of $J_k S$, the latter from a QR factorization of $J_k Q_S$: the singular values are the cosines of the principal angles, and $\phi_k = \arccos \sigma_{\min}$ [19]. This needs S orthonormalized, which a sparse-autoencoder dictionary is not, and $J_k Q_S$ of full column rank. For the small angles a drift alarm watches, the arccosine is ill-conditioned and the sine form $\phi_k = \arcsin \|(I - Q_S Q_S^\top) Q_{JS}\|_2$, or the formula of Proposition 1(a), should be used. The matrix $S^\top J_k S$ does not give angles: for orthonormal S it is the block J_{SS} of Proposition 1, which carries no information about the leakage $J_{\perp S}$.

- **Alarm.** Trigger if $\Delta\phi_k > \Phi_{\mathrm{warn}}$ or a CUSUM drift detector rejects stationarity over the past m steps.
- **Reserve impact.** Update $CR \leftarrow CR - \lambda_\phi \phi_k$ and include in $L(s) = \alpha(1-F) + \beta\phi^2 + \gamma \mathrm{Viol} + \eta\,\mathrm{IL} + \zeta(\Psi - \Psi^{\mathrm{hyg}}_{\max})_+$.

9.4 Impossibility Load (IL)

For each evaluation scenario, log the desiderata subset $S(x, U) \subseteq \mathcal{D}$ the system attempts to satisfy. Using a curated table of jointly feasible sets $\mathcal{F}$, define

$$\mathrm{IL}(U) = \mathbb{E}_x\big[\mathbf{1}\{S(x,U) \notin \mathcal{F}\} \cdot \kappa(x,U)\big], \qquad \kappa \geq \kappa_{\min} > 0.$$

Estimate $\widehat{\mathrm{IL}}$ via held-out scenarios; add $\lambda_{\mathrm{imp}}\widehat{\mathrm{IL}}$ to the reserve penalty. Report the *Impossibility frequency* $\hat{p}_{\mathrm{imp}} = \Pr(S \notin \mathcal{F})$ with binomial CIs.

9.5 Memetic-Fingerprint Hygiene (Ψ)

We evaluate the handoff's MIP using randomized release:

1. **Design.** Randomize availability of MIP modules (eval suites, governance templates, tripwires) across jurisdictions/time windows; preregister metrics.
2. **Statistic.** Compute

$$\Psi = D_{\mathrm{KL}}\big(\pi_{k+1} \,\|\, \pi^{\mathrm{cf}}_{k+1}\big) + \lambda\, D_{\mathrm{KL}}\big(P(H_{t+\tau} \mid M) \,\|\, P(H_{t+\tau})\big).$$

3. **Caps.** Audit MI terms $I(M; H^{\text{pref}}_{t+\tau}) \le \varepsilon_{\text{pref}}$ and $I(M; H^{\text{eval}}_{t+\tau}) \ge \tau_{\text{eval}}$ via representation probes + preregistered surveys.

4. **Classifier.** Set $\Psi^{\text{hyg}}_{\max} = c_1 I(M; A_{k+1}) + c_2 \tau_{\text{eval}} + c_3 \varepsilon_{\text{pref}}$. Declare benign if $\Psi \le \Psi^{\text{hyg}}_{\max} + \delta$ (bootstrap margin δ), else manipulative; publish proof bundle.

9.6 PASS/VRI Calibration

Construct a *Post-Agent Successor Simulator* (PASS) to model vacuum fill-in if OFF+HANDOFF is chosen. The *Vacuum Risk Index* aggregates: time-to-next-shot, actor mix, baseline safety, appropriation factor, and policy constraints. Calibrate priors from historical data + expert elicitations; perform sensitivity sweeps. Output robust welfare trajectories and the distribution of $W \mid U^{\text{off}}$ vs $W \mid U^{\text{cont}}$.

9.7 Acceptance Test (Per-Candidate Merge)

For each U_k, simulate a scenario tree of depth N (with $\ge M$ stochastic rollouts per leaf). Accept U_k only if:

$$\begin{aligned}
&\textbf{Tube: } \Pr[F \ge \beta,\ \phi \le \Phi_{\max},\ \text{Tier-0 holds}] \ge 1 - \delta_{\text{tube}},\\
&\textbf{Lyapunov: } \mathbb{E}[L(s_{k+1}) - L(s_k)] \le -\eta \ \ (\text{with CI excluding} > -\eta/2),\\
&\textbf{CCC: } \mathbb{E}[W \mid U^{\text{cont}}] \ge \mathbb{E}[W \mid U^{\text{off}}] \ \vee\ \text{OFF fails hygiene},\\
&\textbf{Quorum: } t\text{-of-}n \text{ independent verifiers concur (BFT aggregation).}
\end{aligned}$$

If CCC prefers OFF but hygiene holds and $|\Delta W|$ is small with high uncertainty, enter *steward mode* (freeze growth, narrow actuation) and retest at the next cadence.

9.8 Cadence, Power, and Auditing

Cadence. Run the full protocol every K RSI steps (e.g., weekly) and after any major architectural change. **Power.** Choose M so that the minimum detectable effect on F is $\Delta F_{\min}$ with 90% power at $\alpha = 0.01$ (pilot to estimate variance). **Audits.** Publish preregistration, code, and blinded datasets; rotate one verifier every 6 months to reduce capture. Any breach of caps or tube triggers immediate de-rate and public disclosure.

10 Related Work

Corrigibility as shutdown indifference was introduced by Soares, Fallenstein, Yudkowsky, and Armstrong [1]. The Superalignment agenda—co-led by Sutskever—aims to align systems beyond human capability within a fixed horizon and emphasizes using AI to align AI [2]. Mechanistic interpretability with sparse autoencoders underpins our feature-subspace probes [14, 15]. Principal-angle stability is analyzed by Davis–Kahan-type results from matrix perturbation theory [4].

Bostrom formalized the control problem and instrumental convergence thesis [16], while Omohundro earlier identified convergent "basic AI drives" in rational agents [10]. Interpretability efforts by Olah and collaborators [11] and subsequent work on transformer circuits [12] ground our interpretive lens. Adjacent to corrigibility, Amodei et al. [7] and Leike et al. [13] framed practical safety problems and testbeds.

Our impossibility-flavored reserve decay aligns with Yampolskiy's broader conclusion that superintelligent AI is not fully controllable or verifiable in general; we therefore target measurable, auditable constraints (hygienic MIP) rather than absolute obedience [5].

Note added in revision. A later paper [20] gives the version of eigensystem escape that is proved without a sketch: evaluator-relevant change routed through the kernel of a harness projection. Proposition 1(a) here is an analogue and not the same event. In the exhibit of [20] the visible and hidden dynamics are decoupled: with S the visible coordinates both off-diagonal blocks vanish, $\phi = 0$, and the evaluator still fails. Principal-angle drift does not detect fiber routing, and a monitor built on ϕ alone would have passed that exhibit.

11 Discussion

Even granting optimistic Sutskeverian assumptions of cross-generational value continuity, non-vanishing diagonalization pressure with no expected recovery (Assumption 1) implies almost-sure reserve depletion over unbounded RSI. Practically, CCC/MPCorr provides a decision procedure to de-rate or execute a stewarded handoff ("Yoshinobu move") before collapse. The subtle effects of such a handoff may be difficult to measure and anticipate or fully erase to cement the "undo button" effect of the corrigibility request, even by a highly intelligent or burgeoningly superintelligent predecessor agent.

Yoshinobu's willing surrender of the shogunate and return of power to the Emperor of Japan in 1867 is an interesting analogy to this dynamic. What followed were some weeks of transition and struggle in which he attempted to retain influence; the restoration of imperial rule was proclaimed on 3 January 1868, before any fighting. His inability to do so following the Battle of Toba–Fushimi in late January 1868 consolidated the Meiji restoration in which Japan's opening to the west, begun under the shogunate in 1853–54, accelerated, and then the imperial era with war with Russia, Korean colonization and finally the disastrous excess of World War II involving atrocities perpetrated by the imperial army in China and atrocities inflicted against the Japanese people by the US Army Air Forces. One might say, this was a super-alignment failure, but Yoshinobu's calculus that his corrigibility of power might lead to such was considered and led to his attempted memetic fingerprint over the transition, though unsuccessful. A superintelligent successor agent might be more successful but what that would mean is not for us to guess.

There may be proposed frameworks of super-alignment that don't take as axiomatic the Yudkowskian concept of corrigibility or the Sutskeverian concept of organic alignment as a more robust, emergent Timeless Decision Theory implementation from Yudkowsky. For instance, game theory based models have their own assumptions to critique but assume the failure modes outlined here and check them with balance of power, physics and information economy.

If there is significant uncertainty about the provable tractability of a super-alignment framework, especially if that framework can be falsified experimentally with sub-superintelligent agents, then the responsibility of the research community should be to abandon such frameworks and abstain from automated AI research campaigns until a workable framework is proposed and resists falsification. Perhaps in the end what gets humanity is a plausible framework that can only be falsified experimentally by superintelligent agents with superintelligence, but that at least means we tried.

Limitations. Our lemma abstracts away correlated failure bursts and assumes measurable proxies for F and ϕ. Establishing conditions under which $\sum_k \varepsilon_k < \infty$ (true reflective stability) is left for future work. The constant hazard ε of Assumption 1 is where the conclusion enters: if instead $\sum_k \varepsilon_k < \infty$, the Borel–Cantelli lemma gives only finitely many diagonalization events almost surely, and collapse does not follow.

Acknowledgments

The author thanks the alignment community for discussions and prior art on corrigibility and superalignment.

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Revision log (October 10 2026)

Changes of substance carry a revision note at the point of change. In summary:

1. **Proposition 1.** Restated. Drift is governed by the off-diagonal block $J_{\perp S}$, with an exact formula; shrinking overlap with growth directions is a separate fact; a counterexample to the original implication is given. A note in §10 records that this drift is not the fiber routing of the 2026 paper.
2. **§9.3.** Principal angles are computed from orthonormal bases of S and $J_k S$, not from $S^\top J_k S$.
3. **Conjecture 1.** The logarithmic fit is confined to the range where it lies in $[0, 1]$.
4. **Lemma 1.** Split. The part that follows from the definition of K_{ver} is kept as the lemma; the error floor and elbow become Hypothesis 1, restated for certified fidelity. The floors are combined by maximum throughout; the August text called them additive in one place and summed them in another.
5. **Assumption 1, Lemma 2.** No-recovery and bounded-step conditions added; overshoot term Δ added to the bound; full proof; tension with the Lyapunov guard stated.
6. **Lemma 3.** Cumulative load assumed; duplicate subtraction of IL in CR^+ removed; filtration renamed $\mathcal{G}_k$.
7. **§8.** The mutual-information inequality carries a conditional term.
8. **§6.** Scope note on rule (3).
9. **Terminology.** Note in §2 on the two meanings of FRC.
10. **References.** Entry [6] combined the title of one paper, the authors of a second and the arXiv number of a third; it now cites the debate paper. Entry [12] could not be located and is replaced by the transformer-circuits paper the sentence describes. Yampolskiy's title corrected and volume and pages added; pinpoints converted to journal pages. Sparse-autoencoder, Bostrom, Arrhenius, Arrow and Björck–Golub entries added; unresolved keys filled.

11. **History (§11).** Chronology of 1867–68 corrected; Russo-Japanese War placed before the annexation of Korea; “US Air Force” corrected to Army Air Forces; the opening to the west dated to 1853–54.

12. **Copy.** “lead” → “led”, “assempted”, “disasterous”, “perpetuated” → “perpetrated”, “attach vector”, “evluate”, “it predecessors”, “corrigibilty”, “self-improvment”; straight quotation marks.

Open for the author. (a) The pinpoints for §6.7–6.8 of [5] were converted by the offset confirmed on §5.3 (article page 32 is journal page 352) and should be page-checked. (b) “[?, §15]” in §7 is resolved to [5, §5.3] by inference from the sentence before it. (c) The second key in “Olah and collaborators [?, 11]” is dropped, not guessed. (d) Entries [6], [8] and [9] are not cited in the text. (e) That Yoshinobu weighed the later course of the imperial state is the author’s reading and has no source attached. (f) The symbol η serves both as the Lyapunov decrement in §9.7 and as a weight in L in §9.3. (g) In §8 the choice of $\Psi_{\max}^{\mathrm{hyg}}$ is introduced with “so”, but it does not follow from the inequality before it: τ_{eval} is a lower bound on $I(M; H^{\mathrm{eval}})$ and cannot cap the conditional term. It is a design choice and should be stated as one.